

Yulin (Jason) Liu

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SUMMARY

I am a mechanically trained engineer seeking full-time roles in robotics, automation, and systems engineering, with experience applying AI to real-world problems. I focus on building real systems that work under real constraints, turning ideas into reliable solutions across controls and manufacturing environments. I approach problems at the system level to identify failure points, improve performance, and deliver results from design through validation.

PROJECTS

- Biomechanical Culture System** [Cellxercise](#) | *SMAC Integrator* Jan. 2025 – Present
- **Bioreactor Design:** Developed a system delivering 20 N at 10% cyclic strain and 1 Hz to replicate physiologically relevant tensile conditions for tissue growth.
 - **Control System:** Implemented closed-loop LabVIEW control with SMAC actuator and ATO load cell, yielding 0.01 mm positioning resolution and 5 μ m measurement accuracy.
 - **Modular Fabrication:** Fabricated sterilizable steel clamps and aluminum lift plates using MasterCAM, HAAS CNC mills, and WARDJet waterjet.
- Autonomous Vehicles** [TranquiBot](#) | *Project Lead* Jan. 2024 – Present
- **Embedded Systems Architecture:** Architected an autonomous vehicle platform on Jetson Nano, integrating OAK-D depth camera, LD06 LiDAR, and Point One Fusion Engine GPS within ROS2 and the DonkeyCar framework.
 - **Machine Learning & Perception:** Implemented OpenCV-based perception and TensorFlow inference for lane detection and waypoint generation, enabling real-time autonomous navigation from multimodal sensor inputs.
 - **LLM-Driven Autonomy & Control:** Orchestrated ROS2 and Python middleware leveraging the ChatGPT API for high-level decision-making, translating perception outputs into motion commands executed via a VESC motor controller.
- Robotic Lift Mechanism** | *Linkage Designer* Sept. 2022 – Dec. 2022
- **Design & Prototyping:** Generated SolidWorks models and laser-cut prototypes to validate four-bar lift kinematics.
 - **Torque Optimization:** Refined a 1:4 double-gear system to boost lift capacity by 30%.
 - **GD&T Analysis:** Performed GD&T evaluations and drafted detailed force, torque, and kinematic reports.

EXPERIENCE

- Trane Technologies** July 2024 – Dec. 2024
Advanced Manufacturing Engineering Automation Co-op Pueblo, CO
- **Automated Chiller Line:** Led design and commissioning of a turnkey robotic assembly line for shell-and-tube heat exchanger subassemblies, coordinating RFQs and vendor integration to meet production targets.
 - **Standardized Work:** Optimized station layouts and workflows with operators, reducing cycle time by 20% under lean manufacturing guidelines.
 - **Robotic Integration:** Configured FANUC M-710iC 12L robots with Allen-Bradley PLC/HMI, achieving 200 mm/s motion profiles and ± 0.05 mm repeatability.

RESEARCH

- Boomerang Research Laboratory** | *Researcher* Feb. 2023 – Jun. 2024
- **Paper & Simulation:** Contributed to an AIAA paper on wing-tip deflection and built MATLAB airflow models with UWB tracking algorithms.
 - **Experimental Validation:** Conducted 15+ field trials at six joint angles in low-wind conditions, comparing experimental results with simulations to assess the impact of joint angles.
- Xtreme Materials Laboratory** | *Researcher* July 2022 – Jan. 2023
- **Composite Synthesis:** Synthesized high-purity composites via solvothermal methods and ± 10 μ m precision machining, increasing yield by 15%.
 - **Purification Protocols:** Executed acid-washing, ultrasonication, and centrifugation, achieving 93% purity in a controlled nitrogen glove box environment.

LEADERSHIP

- ASME UCSD Chapter** | *Co-Chair* Sept. 2023 – Jun. 2025
- **Operations & Engagement:** Scaled and onboarded a 50-member chapter, leading mentorship programs, technical workshops, and networking events to improve collaboration and member experience.

TECHNICAL SKILLS

Design & Manufacturing: SolidWorks, Fusion 360, CNC, 3D Printing, MasterCAM, PLC (Siemens, Allen-Bradley), LabVIEW, ROS, Closed-Loop Control (PID, System Integration), Python, C++, SQL, Linux, OpenCV, MATLAB, Simulink, ANSYS (FEA)

Additional Background: English, Mandarin, Cantonese, Clarinet (Kansas All-State), Rock Vocalist & Founder, Soccer

EDUCATION

- University of California San Diego** Sept. 2021 – Jun. 2025
B.S., Mechanical Engineering (Control & Robotics) San Diego, CA